

FREEDOM INTERNATIONAL SCHOOL
COMPUTER SCIENCE 2026-27
CSV File Worksheet

CS Board 2023 Main Paper

- i. Write one difference between CSV and text files.
- ii. Write a program in Python that defines and calls the following user defined functions:
 - a) COURIER_ADD ()- It takes the values from the users and adds the details to a csv file 'COURIER.CSV'. Each record consists of a list with the field elements as c_id, sname, source, destination to store courier id, sender name, source and destination address respectively.
 - b) COURIER_SEARCH ()- Takes the destination as input and displays all the courier records going to that destination.

OR

- i) Why is it important to close a file before exiting?
- ii) Write a program in Python that defines and calls the following user defined functions:
 - a) ADD_BOOK ()- Takes the details of the books and adds them to a csv file 'BOOK.CSV'. Each record consists of a list with field elements as book_id, book_name and publisher to store book id, book name and publisher name.
 - b) Search_Book ()- Takes publisher name as input and counts and displays number of books published by them.

CS Board 2023 Supplementary Paper

- i) Write a point of difference between append (a) and write (w) modes in a text file.
- ii) Write a program in Python that defines and calls the following user defined functions :
 - a) Add_Teacher() : It accepts the values from the user and inserts record consists of a list with field elements as T_id, Tname and desig to store teacher ID, teacher name and designation respectively.
 - b) Search_Teacher() : To display the records of all the PGT (designation) teachers.

OR

- i) Write one point of difference between seek() and tell() functions in file handling.
- ii) Write a program in Python that defines and calls the following user defined functions :
 - a) Add_Device() : The function accepts and add records that the consists of a list with field elements as P_id, P_name and Price to store peripheral device ID, device name, and price respectively.
 - b) Count_Device() : To count and display number of peripheral devices, whose price is less than < 1000.

CS Board 2023 Sample Paper

- i) What is the advantage of using a csv file for permanent storage?
- ii) Write a program in Python that defines and calls the following user defined functions:
 - a) ADD() – To accept and add data of an employee to a CSV file 'RECORD.CSV'. Each record consists of a list with field elements as empid, name and salary to store employee id, employee name and employee salary respectively.
 - b) COUNTR() – To count the number of records present in the CSV file named 'record.csv'.

OR

- i) Give any one point of difference between a binary file and a csv file.
- ii) Write a program in Python that defines and calls the following user defined functions:
 - a) add() – To accept and add data of furniture to a CSV file 'FURDATA.CSV'. Each record consists of a list with field elements as fid, fname and fprice to store furniture id, furniture name and furniture price respectively.
 - b) search()- To display the records of the furniture whose price is more than 10000.

CS Board 2023 Sahodva Paper

- i) Sharadhi Patel of Grade 12 is writing a program to create a CSV file 'emp.csv' which will contain employee code, name and salary of employees. Write a program in Python that defines and calls the following user defined function:
 - a) Insrec()- to accept and add data of an employee to a csv file 'emp.csv'. Each record consists of a list with field elements as eid, ename and sal.
 - b) Search()- to display details of all employees whose salary is greater than 100000.As a programmer, help him to successfully execute the given task.

OR

- i) Abhisar is making a software on 'Countries and their capitals' in which various records are to be stored/ retrieved in 'capital.csv' data file. Write a Python program that defines and calls the following user defined functions:
 - a) AddNewRec(Country, Capital)- a function to read and add a new record in csv file 'capital.csv'. Each record consists of the name of the country and its capital as the fields.
 - b) ShowRec()- a function to count and display all records from the CSV file.

CS Board 2024 Main Paper

Sangeeta is a Python programmer working in a computer hardware company. She has to maintain the records of the peripheral devices. She created a csv file named 'PERIPHERALS.CSV', to store the details. The structure of the file is: [Pid, Pname, Price] where:

- Pid is peripheral device (integer)
- Pname is peripheral name (string)
- Price is peripheral device price (integer)

Sangeeta wants to write the following user defined functions:

- a) AddDevice () – to accept a record from the user and add it to the file.
- b) Count_Device () – to count and display the number of peripheral devices whose price is less than 1000.

CS Board 2024 Supplementary Paper

Mr. Mahesh is a Python Programmer working in a school. He has to maintain the records of the sports students. He has created a csv file named 'SPORTS.CSV', to store the details. The structure of the file is: [sport_id, competition, prize_won] where

- sport_id, is Sport id (integer)
- competition is competition name (string)
- prize_won is ("Gold", "Silver", "Bronze")

Mr. Mahesh wants to write the following user-defined functions:

- a) Add_detail (): to accept the details of a student and add to the file.
- b) Count_Medal (): to display the name of competitions in which students have won "Gold" medal.

Help him in writing the code of both the functions.

CS Board 2024 Sample Paper

Vedansh is a Python programmer working in a school. For the Annual Sports Event, he has created a csv file named 'RESULTS.CSV', to store the results of the students in different sports events. The structure of the file is : [St_Id, St_Name, Game_Name, Result]

Where

- St_Id is Student ID (integer)
- ST_name is Student Name (string)
- Game_Name is name of game in which student is participating(string)
- Result is result of the game whose value can be either 'Won', 'Lost' or 'Tie'

For efficiently maintaining data of the event, Vedansh wants to write the following user defined functions:

- a) Accept() – to accept a record from the user and add it to the file. The column headings should also be added on top of the csv file.
 - b) wonCount() – to count the number of students who have won any event.
- As a Python expert, help him complete the task.

CS Board 2024 Sahodya Paper

Raju is a Python programmer working in a leading software company in Chennai. He is assigned with a task to provide the employee name for their annual appreciation with incentive. For this he created a csv file named 'incentive.csv' to store the names of the employees in different criteria. The structure of the file is: [Emp_id, Emp_name, Designation, No_of_Incentives_Received]

He needs to write a user defined functions that perform the following tasks:

- a) Accept()- to accept the details from the user and add it to the file.
- b) Incentive()- to display the name of employee who received incentive more than 3 times.

The file should have proper headers.

CS Board 2025 Main Paper

A csv file 'P_RECORD.CSV' contains the records of patients in a hospital. Each record of the file contains the following data:

- Name of the patient
- Disease
- Number of days patient is admitted and amount.

For example, a sample record from the file is:

['Gunjan', 'Jaundice', 4, 15000]

Write the following Python functions to perform the specified operations on this file:

- a) Write a function read_data() which reads all the data from the file and display details of the 'cancer' patients.
- b) Write a function count_rec() which counts and returns the number of records in the file.

CS Board 2025 Supplementary Paper

Suman is an intern at a software startup. The company has assigned her a task to create a CSV file named 'CLUB.CSV', to store the records of the Club members. After discussing with Club Incharge, Suman has planned to store the following content of members in the file CLUB.CSV :
[Mno, Name, Mobile, Fee]

Where

- Mno – Member Number
- Name – Name of the Member
- Mobile – Member's Mobile Number
- Fee –Fee amount.

Assuming you are asked to help Suman in her assignment, write a Python code for performing the following tasks with the help of user-defined functions :

- a) NewMembers() : to accept records of members from the user and add them to the file CLUB.CSV.
- b) PriorityMember() : to find and display those members from the file CLUB.CSV, who are paying a fee of more than 35000.

CS Board 2025 Sample Paper

A csv file "HAPPINESS.CSV" contains the data of a survey. Each record of the file contains the following data:

- Name of a country
- Population of the country
- Sample Size (Number of persons who participated in the survey in that country)
- Happy (Number of persons who accepted that they were Happy)

For example, a sample record of the file may be: ['Signiland', 5673000, 5000, 3426]

Write the following Python functions to perform the specified operations on this file:

- i) Read all the data from the file in the form of a list and display all those records for which the population is more than 5000000.
- ii) Count the number of records in the file.

CS Board 2025 Sahodva Paper

Write a program in Python that defines and calls the following user defined functions:

- i) ADD() – To accept and add data of a room booked to a csv file 'HOTEL.CSV'. Each record of the file is as follows [Room_no, Room_Type, Price, Number_of_guest]. Room_no, Room_Type are of string type, price and number_of_guest is of integer type.
- ii) Display() – To read and display only those data from the file whose Room_type is "Delux". Also count and return the total number of guests and total amount collected for "Delux" room.

OR

Vijay has created a CSV file 'FOOTBALL.CSV' to store player details, like

- player name
- matches played
- goals scored.

The file has the following columns as headers: [PNAME, MATCH, GOALS].

Write the following user-defined functions in Python:

- i) addPlayer(name, matches, goals): This function takes three parameters. It should create a list containing these details and append the data to the existing CSV file FOOTBALL.CSV without overwriting previous entries.
- ii) HighestAverage(): This function reads FOOTBALL.CSV and displays the name of the player with the highest average goals per match (i.e., goals/matches). Ensure that the function does not calculate average where matches might be zero.

CS Board 2026 Main Paper

A csv file "STATES.CSV" contains some data about all the states of India. Each record of the file contains the following data :

- Name of the State
- Capital of the State
- Population of the State
- Official Language of the State

For example, a sample record in the file is : ['Andhra Pradesh', 'Amaravati', 52221000, 'Telugu']

Write a Python program which reads the data from this file and appends all those records where population is more than 10000000 into another csv file 'MORE.CSV'.

Note : "STATES.CSV" also contains the Header row. The Header row should NOT be copied to "MORE.CSV".

CS Board 2026 Sample Paper

Raj is the manager of a medical store. To keep track of sales records, he has created a CSV file named 'SALES.CSV', which stores the details of each sale. The columns of the CSV file are:

- Product_ID
- Product_Name
- Quantity_Sold
- Price_Per_Unit.

Help him to efficiently maintain the data by creating the following user-defined functions:

- i) Accept() – to accept a sales record from the user and add it to the file.
- ii) CalculateTotalSales() – to calculate and return the total sales based on the Quantity_Sold and Price_Per_Unit.

CS Board 2026 Sahodva Paper

Sanya runs a stationary shop and maintains her daily transaction details in a csv file named 'stock.csv'. The file stores details in the format: [Itemid, Itemname, Quantity, Cost_per_item]

Write user defined Python functions to help her manage the stock:

- a) AddItem()- To accept an item record from the user and append it to the file. Validate that quantity>0 and cost_per_item>0 before storing the details.
- b) TotalCost()- to read the data from the file and calculate and return the total cost of all items in stock by multiplying quantity*cost_per_item for each record.

Practice Questions:

1. Amit runs a book store and keeps his daily sales records in a csv file 'BOOKS.CSV'. The file stores the following columns: Book_Id, Title, Author, Copies Sold and Price_Per_Copy. Write user defined functions to perform the following tasks:
 - i) Add_Record()- to accept a book's sales records from the user and add it to the file
 - ii) HighestEarningBook()- to read data from the file and display title of the book that generated maximum total revenue (where revenue= Copies_Sold*Price_Per_Copy)
2. Create a function maxsalary() in Python to read all the records from an already existing file record.csv which stores the records of various employees working in a department. Data is stored under various fields as shown below:

E_code	E_name	Scale	Salary
A01	Bijesh Mehra	S4	65400
B02	Vikram Goel	S3	60000
C09	Suraj Mehta	S2	45300
.....			

Function should display the row where the salary is maximum. Note: Assume that all employees have distinct salary.

3. Write a Python program to create a csv file 'DVD.CSV' and write 10 records in it Dvd_id, dvd name, qty and price.
Write a function to display those dvd details whose dvd price is more than 25.
4. Aman has recently been given a task on the file 'CAPITAL.CSV', which contains Country and Capital as data.
For Example: "INDIA", "NEW DELHI" "CHINA", "BEIJING"
Write a Python code for these two user defined functions:
 - i) AddNewRecord(): insertion of new records in file.
 - ii) ShowRecord() : to display all the records from file.
5. A CSV file "HealthData.csv" contains the data of a health survey. Each record of the file contains the following data:
Name of a country
Life Expectancy (average number of years a person is expected to live)

GDP per capita (Gross Domestic Product per person)

Percentage of population with access to healthcare

For example, a sample record of the file may be: ['Wonderland', 82.5, 40000, 95].

Write the following Python functions to perform the specified operations on this file:

- i) Read all the data from the file in the form of a list and display all those records for which the life expectancy is greater than 75.
- ii) Count the number of records in the file.